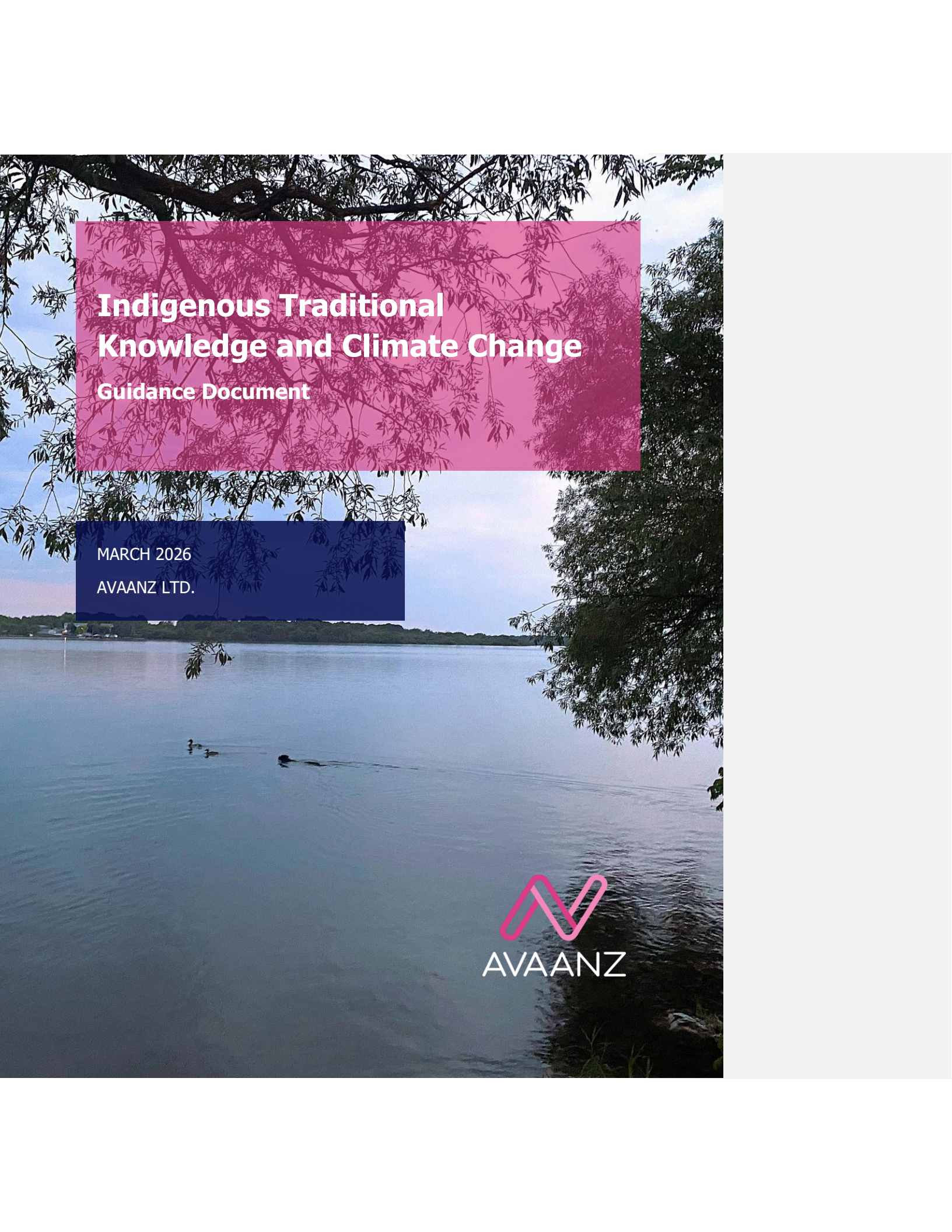


The background of the cover is a photograph of a calm lake. In the foreground, there are dark, leafy tree branches hanging down from the top and right sides. The water is a deep blue-grey color. In the distance, a line of green trees marks the opposite shore under a pale, overcast sky. A semi-transparent pink rectangle is positioned in the upper left, containing the title. Below it, a smaller dark blue rectangle contains the date and company name.

Indigenous Traditional Knowledge and Climate Change

Guidance Document

MARCH 2026
AVAANZ LTD.



Indigenous Traditional Knowledge and Climate Change Guidance Document



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1. Acknowledgments

The project team is grateful to the Indigenous community reviewers who took the time to meet and share their perspectives on the document and the topic. Their insightful feedback has strengthened the guidance document.

2. Introduction

This document has been developed based on the results of a research project involving a literature review of published peer-reviewed articles and grey literature, as well as feedback from Indigenous community reviewers, and professional and personal experience among the project team. This document does not necessarily reflect the views of the electricity sector.

*This document remains "evergreen". The authors welcome and look forward to ongoing feedback from Indigenous communities and practitioners. Any knowledge and experience that may be shared will help support continuous learning and ongoing improvements to the document. Please send your feedback to ****@****.*

The aim of this document is to assist practitioners (e.g., environmental engineers, specialists, and planners) in the Canadian electricity sector to better understand and include Indigenous knowledge in the development and/or modification of projects in the context of climate change. This includes understanding what Indigenous knowledge is, who owns, controls, accesses and possesses this knowledge, and how it can ensure a more accurate and comprehensive understanding of a project site including the local impacts or indicators of climate change. It is also critical that practitioners acknowledge and understand the history of interactions between Indigenous communities, developers, researchers, and government agencies under colonialism. This context continues to impact relationships and collaboration between Indigenous communities and developers and researchers today.

This document provides practical and adaptable guidance for practitioners to seek, receive, and apply Indigenous knowledge to electricity projects related to climate change largely within but not limited to the environmental assessment context. Based on the source material, language in the guidance document is addressed at times to practitioners, engineers, those in technical positions, researchers, and proponents but understood to be relevant to practitioners throughout. It is understood there are both research and impact/environmental assessment processes that relate to seeking, receiving, and applying Indigenous knowledge that would apply to electricity sector project development/modification. While this document aims to provide actionable guidance, it is important to keep in mind that every Indigenous community and knowledge holder are different. We also heard that communities, their landscapes, Elders, and realities are in a constant state of change. There is no single approach that fits all communities and practitioners must communicate with communities directly to understand what works for each community. The guidance document is organized into the following sections.

Commented [AY1]: As shown, we would be able to include a general email address for feedback if appropriate.

- Acknowledgments
- Introduction
- Background
- Methods
- Indigenous knowledge
 - Definition of Indigenous knowledge
 - Intellectual property
 - Guidelines and policies
- Promising practices for applying Indigenous knowledge
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3. Background

3.1 Climate change and Indigenous knowledge

Climate change is impacting the Canadian electricity sector and is an important consideration when developing and/or modifying electricity projects in Canada. However, western science can present gaps and limitations around climate change data, especially at the level of individual electricity projects. Indigenous knowledge offers an alternative source of knowledge that is complementary to western science (Eckert et al., 2020).

Indigenous knowledge holders can provide important baseline information about the environment as well as the local history, variability, and extent of biophysical, social, and cultural phenomena (Jessen et al., 2021) (British Columbia Environmental Assessment Office, 2022) (Mackenzie Valley Review Board, 2005). As it is a source of multigenerational knowledge for understanding processes within the traditional territory (Jessen et al., 2021), the information provides a more accurate and community-specific narrative of long-term ecological change (Lajoie-O'Malley et al., 2023), which could include more nuanced and site-specific insights into climate change and in the context of extreme cold and extreme heat (Arsenault et al., 2019).

Indigenous knowledge can also help:

- Identify links between parts of the environment that may not seem related (British Columbia Environmental Assessment Office, 2022; Mackenzie Valley Review Board, 2005).
- Predict potential impacts related to a project (British Columbia Environmental Assessment Office, 2022) (Mackenzie Valley Review Board, 2005).
- Make predictions and measurements more accurate (British Columbia Environmental Assessment Office, 2022).

- Help avoid or mitigate impacts to sensitive areas (British Columbia Environmental Assessment Office, 2022).

Together, Indigenous knowledge and western science can produce a more comprehensive understanding (Impact Assessment Agency of Canada, 2024a) that could enhance the climate resilience of electricity projects, as well as support the accurate characterization and mitigation of a project's potential impacts within an assessment process.

"CLIM086 Indigenous Knowledge and Perspectives of Climate Adaptation", an online course offered by Royal Roads University¹, is another resource that speaks to the importance of integrating Indigenous knowledge and western science in the context of climate change adaptation.

3.2 Reconciliation

For practitioners to seek, receive, and apply Indigenous Knowledge, they must first understand the history of colonialism in Canada and how it has shaped relationships between Indigenous communities and government agencies, developers, and researchers. This history includes displacement, land dispossession, and policies of assimilation including residential schools (The Standing Senate Committee on Aboriginal Peoples, 2019).

At a minimum, practitioners who wish to engage with Indigenous knowledge should complete training on the First Nations principles of ownership, control, access, and possession – OCAP®². The OCAP training supplements the information contained in this guidance document. A fulsome exploration of reconciliation, the history of Indigenous communities in Canada and implications for electricity sector practitioners is beyond the scope of this guidance document. However, developing this understanding and meaningfully considering reconciliation within professional practice is necessary to engage with Indigenous communities around Indigenous knowledge. Additional resources to learn more about these topics could include:

- Indigenous Canada [online course offered by the University of Alberta³] (<https://www.coursera.org/learn/indigenous-canada>)
- National Centre for Truth and Reconciliation (<https://nctr.ca/>)
- Truth and Reconciliation Commission of Canada: Calls to Action (https://publications.gc.ca/collections/collection_2015/trc/IR4-8-2015-eng.pdf)
- National Inquiry into Missing and Murdered Indigenous Women and Girls: Calls for Justice (https://www.mmiwg-ffada.ca/wp-content/uploads/2019/06/Calls_for_Justice.pdf)
- The Standing Senate Committee on Aboriginal Peoples: How did we get here? A concise, unvarnished account of the history of the relationship between Indigenous Peoples and

¹ Offered at the time of guidance development

² <https://fnigc.ca/ocap-training/>

³ Free at the time of guidance development

Canada (https://sencanada.ca/content/sen/committee/421/APPA/Reports/APPARReport-Phase1_WEB_e.pdf)

- United Nations Declaration on the Rights of Indigenous Peoples (<https://social.desa.un.org/issues/indigenous-peoples/united-nations-declaration-on-the-rights-of-indigenous-peoples>)
- *Some electricity sector organizations may also provide relevant internal training*

Including Indigenous knowledge in the context of an environmental assessment process not only supports project outcomes such as ensuring climate resilience but supports reconciliation, and recognizes Indigenous governance, as well as jurisdiction and decision-making (British Columbia Environmental Assessment Office, 2022). There is an opportunity to acknowledge and move away from approaches that have been connected to Indigenous Peoples' dispossession from their lands.

4. Methods

This guidance document was developed based on the results of a desktop review and feedback received from Indigenous community reviewers as well as professional and personal experience among the development team. The desktop review focused on relevant peer-reviewed and grey literature, which aimed to identify promising practices and principles related to seeking, receiving, and applying Indigenous knowledge to the development and modification of projects in the Canadian electricity sector within the context of climate change. Community-specific guidelines or policies were also sought out. The Google and Google Scholar search engines were used to identify relevant articles. The search focused on Indigenous knowledge, Canadian sources with relevance to the electricity sector, First Nations or Métis communities, and developers/practitioners. Screened-in sources were published within the past ten years or identified specifically by project collaborators. Although there was an emphasis on environment or impact assessment, peer-reviewed articles related to Indigenous knowledge within a research context were also included. The applications of Indigenous knowledge can be broad and not limited to impact/environmental assessment at a project site.

Examples of key words used in the searches included:

- | | |
|-------------------------|-----------------------|
| • Indigenous knowledge | • hydropower project |
| • Traditional knowledge | • electricity |
| • Aboriginal knowledge | • development |
| • local knowledge | • renewal |
| • Canada | • promising practices |
| • climate change | • wise practices |
| • energy | • best practices |
| • power | |

Due to the large number of peer-reviewed journal articles retrieved from the Google Scholar search, the articles were organized into "highly relevant" or "less relevant" groups, with a focus on

drawing more heavily from “highly relevant” sources to develop the guidance document. This content was supplemented by the less relevant articles. The screening criteria are outlined below.

Highly Relevant articles:

- Written by or collaborated with an Indigenous author (First Nations, Métis)⁴.
- Written with an engineering or energy sector lens.
- Included discussion, definition of, or nation-specific description of Indigenous knowledge.
- Included advice for technical positions.
- Included guidance, plans of action, or policies from and for First Nations or Métis communities.
- Included guidance on seeking Indigenous knowledge within environmental assessments / impact assessments, specifically.
- Included guidance on applying Indigenous knowledge within environmental assessments / impact assessments, specifically.

Less Relevant/Supplemental articles:

- Written by a non-Indigenous author.
- Definition(s) of Indigenous knowledge, Indigenous knowledge.
- Community member opinion-based information.
- Guidance on implementing Indigenous knowledge within energy sector/projects, but not environmental assessment / impact assessment specific.

A Google search was also undertaken to identify community-specific guidelines and policies shown in Section 5.3. A further Google Scholar search was carried out to identify examples of projects that implemented the “Two-Eyed Seeing” approach to integrate Indigenous knowledge alongside western science to provide insights around the process.

5. Indigenous knowledge

5.1 Definition of Indigenous knowledge

There is no single universally accepted definition of Indigenous knowledge (Impact Assessment Agency of Canada, 2024a). The knowledge is diverse and can be described in many ways (British Columbia Environmental Assessment Office, 2022). Indigenous knowledge is often defined as a body of place-based knowledges accumulated and transmitted across generations within specific cultural contexts (Jessen et al., 2021). Indigenous knowledge is defined by the place-based expertise continued to be depended on and passed down intergenerationally today, as it is rich in culture and involves multigenerational relationships with lands and waters as a result of Indigenous people continuously adapting their ancestral knowledge to the unexpected changes of

⁴ Articles may also cite other scholars including Indigenous scholars. In the interest of ensuring the document is streamlined to support its use, secondary sources have not necessarily been identified but can be readily accessed through the main reference.

local seasonal cycles (Turner & Reid, n.d.). This knowledge is broad, holistic, and rooted in the worldview of Indigenous people (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a). Indigenous knowledge involves a deep respect of cultural and spiritual knowledge that the local human-environmental relationships depend on (Jessen et al., 2021).

Knowledge holders accumulate information in ways supported and supplemented by teachings from oral histories and cultural practices, including harvesting, observation, animal husbandry, and experimentation (Jessen et al., 2021). Ancestors of Indigenous communities developed place-based methods for tracking the changing of seasons and ideal times for activities through the naming of different moons, tracking the flowering certain plants, or tracking the songs of certain birds (Turner & Reid, n.d.). Indigenous communities' cultural activities or practices vary from region to region based on the local conditions, their knowledge of the environment, and local weather patterns such as rainfall, snow, wind directions, and summer heat (Turner & Reid, n.d.). Within Indigenous communities, families and clan groupings continue to have their own recognized places for harvesting and processing food, materials, and various medicine species (Turner & Reid, n.d.). Knowledge may be generated, transmitted, and held in different ways within a community depending on gender, age, and cultural roles, creating specialists on certain topics among community members (Jessen et al., 2021).

Indigenous communities continue to apply Indigenous knowledge and experiences into planning, decision-making, seasonal movements, and seasonal harvesting (Turner & Reid, n.d.). Knowledge holders have noted that year-to-year differences are possible, depending on conditions such as weather or ocean currents, and the knowledge systems developed and fostered over generations have allowed communities to respond to unanticipated cycles of environmental change (Turner & Reid, n.d.).

Indigenous traditional knowledge is also referred to as "traditional knowledge" or "Indigenous knowledge" (Dimayuga et al., 2020), and "traditional ecological knowledge" or "Indigenous ecological knowledge" when the knowledge is related to ecology (Jessen et al., 2021). Indigenous traditional knowledge and the accompanying terms represent the collective knowledge of traditions used by Indigenous groups to sustain themselves to their environment over time, passed on from one generation to the next (Dimayuga et al., 2020). Although the language "traditional" suggests a historical aspect, this knowledge continues to evolve (Impact Assessment Agency of Canada, 2024a). For consistency and based on the literature review results where articles primarily used the term "Indigenous knowledge" over "traditional knowledge" or "ecological knowledge", "Indigenous knowledge" has been used through this guidance document as an umbrella term for these related concepts.

Indigenous knowledge can take various forms. For example, it could be documented orally, in writing, in photos, videos, and reflected through ceremonies or traditional activities (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a). In an assessment process, Indigenous knowledge could include biophysical observations or

experiences, information about practices, rituals, teachings, laws, language, and relationships as well the values of the community (British Columbia Environmental Assessment Office, 2022).

5.2 Intellectual property

Any Indigenous knowledge shared is the intellectual property of the knowledge holders and their communities, as these knowledge systems are inseparable from the people who carry them and their traditional territories (Turner & Reid, n.d.). All Indigenous knowledge utilized throughout the project lifetime must be directly attributed to the community source to protect ownership and uphold data sovereignty (Dimayuga et al., 2020). Practitioners should understand and apply the First Nations principles of ownership, control, access, and possession – OCAP⁵ to ensure they respect and support Indigenous communities' ownership, control, access, and possession of their Indigenous knowledge. Practitioners should also understand and respect Article 31 of the United Nations Declaration on the Rights of Indigenous Peoples which protects the Indigenous right "to maintain, control, protect, and develop their cultural heritage, traditional knowledge, and traditional cultural expressions" (Bruno et al., 2025).

See Section 6.2.1.6 for detailed discussion around confidentiality and consent when seeking and receiving Indigenous knowledge.

5.3 Guidelines and policies

A number of Indigenous communities have developed community-specific policies, protocols, and other guidance related to Indigenous knowledge, as well as research and consultation more broadly. Community-specific policies, protocols, and principles are locally-developed guidelines for culturally appropriate research with First Nations and Métis communities (Dimayuga et al., 2020); they can also contribute to forming the co-management or co-creation structures within an Indigenous-led environmental assessment (Eckert et al., 2020).

Locally-developed guidelines allow for all parties involved to ensure the project progresses aligned with the teachings of the territory, such as First Nations in the Manitoulin Island area structuring policies to be aligned with the Anishinaabe Seven Grandfather Teachings of respect, wisdom, love, honesty, humility, bravery, and truth (Dimayuga et al., 2020). On the Pacific coast, the Haítzaqv designed and implemented a community-centred grizzly bear monitoring program guided by the customary law of Gvi'ilas, an approach that combined cultural values with western knowledge of bears, salmon, and people on the central coast used within the development of a contemporary wildlife management policy (Jessen et al., 2021).

As described in Section 6.2, such guidance should be reviewed before engaging with Indigenous communities where available and otherwise as soon as they can be accessed in the process of

⁵ <https://fnigc.ca/ocap-training/>

seeking Indigenous knowledge. Examples of policies, protocols, and principles are shown below. These documents address topics such as:

- Principles related to the use and management of Indigenous knowledge
- Access
- Consent to use
- Requesting access
- Confidentiality requirements
- Rights, including intellectual property rights
- Maintaining integrity
- Consultation
- Related protocols, procedures, processes, and policies including steps before, during, and after research
- Agreements
- Roles and responsibilities

5.3.1 First Nation of Na-Cho Nyak Dun Traditional Knowledge Policy (2008), Yukon

Source: <https://ethicshub.ca/wp-content/uploads/2020/05/75-First-Nation-Protocol-For-Conducting-Traditional-Knowledge-Research-or-Accessing-Traditional-Knowledge.pdf>

5.3.2 Sambaa K'e Dene Band Policy Regarding the Gathering, Use, and Distribution Of Yúndíit'ōh (Traditional Knowledge) (2003), Northwest Territories

Source: <https://nwtresearch.com/sites/default/files/sambaa-k-e-dene-band.pdf>

5.3.3 First Nations in Quebec and Labrador's Research Protocol (2014), Quebec, Labrador

Source: https://achh.ca/wp-content/uploads/2018/07/Protocol_FN-Research-Protocol-in-Labrador-and-Quebec.pdf

5.3.4 Principles for Proponents working in the Traditional Territories of the Saugeen Ojibway Nation (n.d.), Ontario

Source: <https://www.saugeenojibwaynation.ca/document/consultation-principles-proponents>

6. Promising practices for applying Indigenous knowledge

6.1 Principles

Seeking, receiving, and applying Indigenous knowledge should be informed by the principles described in the sections below. See also Sections 5.2 and 6.2.2 for additional discussion of principles and approaches related to intellectual property and community ownership, control, access, and possession of Indigenous knowledge.

6.1.1 Respect

Seeking, receiving, and applying Indigenous knowledge must be based on respect for the worldview of Indigenous peoples (Impact Assessment Agency of Canada, 2024a) (British Columbia Environmental Assessment Office, 2022). The continued generation of Indigenous knowledge also relies on Indigenous communities exercising their treaty rights and by extension respect must extend to Indigenous and treaty rights (Impact Assessment Agency of Canada, 2024a).

For research projects, showing respect means understanding the research in relation to the associated people and territory. Within this lens, reciprocity is needed and involves a sustained, genuine effort to make research processes accessible, such as a co-developed research design, sovereign data collection, and adequate funding to ensure the community can fully participate in consultation. Respect extends to ensuring the privacy of Indigenous community members and sensitive information received regarding cultural, spiritual, or ecological knowledge (Jessen et al., 2021). See also Section 6.2.2 for further discussion around consent and confidentiality.

6.1.2 Relationship-based

Relationships with Indigenous communities and knowledge holders are fundamental to seeking, receiving, and applying Indigenous knowledge (British Columbia Environmental Assessment Office, 2022).

In the Nishnaabeg tradition, animals will be acknowledged and included as an equal partner within the project, as Nishnaabeg communities hold diplomatic agreements between human and animal nations within cultural traditions. This is reflected in Nishnaabewin protocols that guide and support the associated practices, knowledges, and ethics that make up their people and construct their Nishnaabeg world. Acknowledging, respecting, and extending the relationship to non-human animals, especially those held in high positions of power within the local clan system, within the Indigenous communities could lead to meaningful discussions of the natural world (Lajoie-O'Malley et al., 2023).

6.1.3 Transparency

Indigenous knowledge must be sought, received, and applied in a way that is fair, and open (British Columbia Environmental Assessment Office, 2022).

6.1.4 Permission

Indigenous knowledge may only be applied with the permission of Indigenous communities and knowledge holders. The community's requirements and approaches must be followed (e.g., laws, policies, practices, governance) (British Columbia Environmental Assessment Office, 2022). See Section 6.2.2 for further discussion of consent around applying Indigenous knowledge.

6.1.5 Acknowledgment of context

Indigenous knowledge is contextual and must be considered in the context within which it was shared (British Columbia Environmental Assessment Office, 2022).

6.1.6 Flexibility

Seeking, receiving, and applying Indigenous knowledge must be done in a way that is flexible, taking into consideration the unique needs and approaches of each Indigenous community (Impact Assessment Agency of Canada, 2024a). Extra consideration must be taken for northern or remote Indigenous communities, as timelines and plans must carefully build in sufficient resources for engagement to navigate the logistical challenges (Dimayuga et al., 2020). It is important to be flexible with the timing and location of any meetings or engagement sessions, as Indigenous community members may require times or locations that work best with their careers or on-the-land activities. Be prepared for interviews or engagement sessions to take place at informal venues, such as the band office, restaurants, friendship centers, or the participants residence, if invited (Belisle et al., 2022).

6.2 Seeking and receiving Indigenous knowledge

6.2.1 Overall approach

6.2.1.1 Build and maintain relationships

From the outset, establish and maintain relationships with Indigenous communities to ensure effective contact and communication in preliminary project stages, throughout the project development, and after project completion or monitoring (Arsenault et al., 2019). Time and effort are needed to build trust and relationships, and it is important to engage with Indigenous communities early in any process (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a, 2024b; Mackenzie Valley Review Board, 2005). Commit to outreach well in advance of the project timeline to introduce yourselves and engage with community early on. Make the time necessary to build a relationship with the Indigenous

community directly. It is not possible to meet the community members and see the lands they are inherently connected to from behind a screen. Personal, face-to-face communication will demonstrate greater respect for the Indigenous community and provide more of an opportunity for real connection than an indirect, passive approach such as surveys or online meetings. The importance of relationships cannot be emphasized enough. Building relationships is not a one-time, superficial activity but instead underpins all other sections in this document.

6.2.1.2 Learn about the Indigenous community

First Nations, Métis, and Inuit all have unique knowledge and each Indigenous community is unique (Impact Assessment Agency of Canada, 2024a). Engage each Indigenous community separately unless communities would like to work together (Impact Assessment Agency of Canada, 2024a). Before engaging with an Indigenous community, it is important to seek and review existing information about the community to better understand the community's context (Impact Assessment Agency of Canada, 2024a). Make an effort to understand their history and traditions (British Columbia Environmental Assessment Office, 2022). Learning about and understanding the history of the Indigenous community demonstrates respect. It is important to understand how the community governs itself and which treaty, if any, the Indigenous community is a signatory to. Communities may reside on lands or waters designated for a different treaty due to the vast traditional territorial reach.

If invited, or encouraged to attend, participate in community events such as fishing derbies, career fairs, or festivals and powwows to meet community members and gain an understanding of their ways of life. When visiting the community, proponents should seek to understand the community's traditional territory including its lands and waters, the community's relationship with its ancestors, the histories that continue to effect reconnection to the land today, and why the community feels so strongly about being involved in discussions and information-sharing around activities that can impact the land and their community.

6.2.1.3 Engage with the Indigenous community

Begin engagement with the Indigenous community through community leadership to ensure it is clear how the community's processes or protocols can be respected (Impact Assessment Agency of Canada, 2024a). Should the community's elected leadership not be available, contact the community's applicable consultation department (Lands & Resources, Rights & Research). Seek permission to enter the community. Approach engagement with cultural humility and a desire to learn together. Do not assume that you have greater knowledge or understanding than the community.

Seek and review any existing information about the community's approach to consultation and sharing knowledge (Impact Assessment Agency of Canada, 2024a) but ask about cultural protocols to follow, about how it would be best to engage, and how to include Indigenous knowledge (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a). This includes best practices around engaging with women, Elders, harvesters, and youth

(Impact Assessment Agency of Canada, 2024a). Also seek to understand how to work with knowledge holders. (Mackenzie Valley Review Board, 2005). Protocols and procedures could be written or oral and could relate to the collection, use, approval, validation, review, interpretation, and protection of Indigenous knowledge (Impact Assessment Agency of Canada, 2024a). Each community will have differing protocols to portray respect, such as gifting tobacco ties, providing the opportunity to smudge, or inviting an Elder to provide an opening and closing prayer. Understand, respect, and adhere to the Indigenous community's policies, protocols, procedures, governance, and laws related to Indigenous knowledge (Impact Assessment Agency of Canada, 2024a; Mackenzie Valley Review Board, 2005). See Sections 5.3 and 6.2.2.3 for further discussion around community-specific policies, protocols, and guidelines.

6.2.1.4 Preparation for practitioners

Consider if team members may benefit from Indigenous cultural training before engaging (British Columbia Environmental Assessment Office, 2022). Collaborative research with Indigenous communities and partners requires engineers and technical staff to recognize the harms research historically imposed on Indigenous people, the ways in which Indigenous knowledge is discounted in application, and the realities of Indigenous knowledge being inappropriately reproduced, applied, or misused (Jessen et al., 2021). See also Section 3.2 for further discussion of practitioners and reconciliation.

6.2.1.5 Focus on the process

Focusing on the process, rather than end products, allows for greater insights through relationship-building between technical staff and Indigenous community members as the focus shifts from technical details and results to respectful, meaningful information sharing (Dimayuga et al., 2020), as well as a greater comprehension of the Indigenous community's relationship to the territory. By focusing on the process, proponents move away from the western approach of efficiency via transactional relationships with Indigenous communities in exchange for connections based in mutual understanding and respect.

6.2.1.6 Shared research priorities

Where knowledge is sought from a research perspective, prior to project initiation and data collection, the proponent and engineering team should endeavour to earn the trust of the Indigenous community members and foster healthy working relationships to identify community-specific research priorities (Jessen et al., 2021). The engineering team can co-develop a research agreement around these research priorities. Research should be developed in a transparent and collaborative manner with the Indigenous community including input from Knowledge holders (Jessen et al., 2021), that demonstrates acknowledgement and respect of both Indigenous and western empirical worldviews (Dimayuga et al., 2020). Engineers can combat the extractive nature practiced within the environmental impact assessment process by inviting local Indigenous community members and leadership to participate in project planning, assessments and decision-making processes, and relationship-building with the project proponents and regulators (Arsenault

et al., 2019). Shared research priorities are an important part of ensuring collaboration and information exchange between the community and researchers/engineers.

The proponent should consider co-developing an Indigenous Stewardship Body with the Indigenous community to further strengthen the Indigenous community's involvement within the research process. Invite trusted, knowledgeable community members to advise the engineering team on seeking or applying Indigenous Knowledge, working with Indigenous Knowledge holders, and the implementation or monitoring of the project (Bruno et al., 2025).

6.2.2 Confidentiality and consent

6.2.2.1 Informed consent

It is important to discuss confidentiality considerations and information management with the Indigenous community before any Indigenous knowledge is shared (Impact Assessment Agency of Canada, 2024b). Discuss with the Indigenous community why they are being asked to provide Indigenous knowledge and how it will be used (Impact Assessment Agency of Canada, 2024a). The knowledge holders must be made aware of how and where the information being shared will be used within the process, who will be analyzing and reviewing the information, and opportunities to review the implementation of their knowledge prior to submission (Turner & Reid, n.d.). Ensure the Indigenous community and community members have ample time to consider a request for their consent and how it could affect their community. Be clear with the Indigenous community about your intentions; some community members may be reluctant to participate in consultation but may be more open to information-sharing.

An example – A cultural heritage project in Ontario engaged First Nation community members verbally and in-writing at the outset of discussions regarding consent to record and share the stories, community information, or any sensitive information community members provided during discussions. Community members were provided ample opportunity to revise and approve of their contributions.

6.2.2.2 Ongoing consent

Sharing Indigenous knowledge can be a sacred activity and it is a privilege when Indigenous communities share Indigenous knowledge in the context of a project (Impact Assessment Agency of Canada, 2024a). Indigenous knowledge can only be used with permission and informed consent from knowledge holders unless it is explicitly public (Mackenzie Valley Review Board, 2005). Seek, and do not assume, consent from Indigenous community leaders and knowledge holders to include Indigenous knowledge in any work before you engage any knowledge holders. Consent needs to be ongoing (Impact Assessment Agency of Canada, 2024a). Ask if the Indigenous knowledge can be recorded in writing; some information should not be written (Impact Assessment Agency of Canada, 2024a). Respect that an Indigenous community may choose not to share Indigenous knowledge (British Columbia Environmental Assessment Office, 2022). While acknowledging the idea of "open data" science, keep community-specific cultural, spiritual, or ecological information confidential when working with Indigenous knowledge as discussed with the community (Jessen et

al., 2021)⁶. Respect the wishes of the community or Knowledge holder by not including any confidential information shared within reports (Turner & Reid, n.d.). Prevent potential identification of participants or sensitive community information by using aggregated results and higher-level analysis when reporting (Almack et al., 2023). If relevant information was gathered for a different purpose and already exists (e.g., a Indigenous knowledge study), seek consent from the Indigenous community to apply the existing information to the new purpose (Impact Assessment Agency of Canada, 2024a).

6.2.2.3 Community policies and protocols

Follow the Indigenous community's protocols or policies around the sharing and use of Indigenous knowledge or if these do not exist, agree on an approach with the Indigenous community for how written and oral information will be kept confidential and handled and how Indigenous knowledge will be included in a project/assessment (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024b). This includes understanding if community members need to have agreements from leadership to share Indigenous knowledge (Impact Assessment Agency of Canada, 2024a). As specific knowledges must be directly attributed to the source to protect ownership, participants may wish to be named within the project records (Dimayuga et al., 2020). The approach to sharing, use, and management of Indigenous knowledge must be documented and formally agreed upon (Impact Assessment Agency of Canada, 2024b). However, approach each community as though they have a distinct form of approving permission to be recorded, as requiring written consent as the only form may be incompatible with the oral traditions of the community being collaborated with (Dimayuga et al., 2020). Collaborate with Indigenous knowledge holders to create tools, such as frameworks, strategies, or workplans to advance the sharing of Indigenous knowledge in a way that acknowledges, promotes, protects, respects, and honours Indigenous knowledge (Bruno et al., 2025).

From a research perspective, initiate the working relationship with the Indigenous community by co-developing a Memoranda of Understanding (MOU) or a Research Agreement that details the agreed upon research project aims, respective responsibilities, data protection, and ownership over the Indigenous knowledge shared (Dimayuga et al., 2020). When co-developing a data-sharing agreement with an Indigenous community, ensure the principles of OCAP[®] are respected and applied to ensure that the Indigenous community maintains full control over their data. The handling of a community's historical, contextual, or shared knowledge should be regulated by a data release agreement and human ethics protocols between the Indigenous community and proponent that follows the principles of OCAP[®] and the collective benefit, authority to control responsibility, ethics (CARE) for Indigenous People's Data Governance (Galatro et al., 2024). See also Section 5.2 for further discussion of intellectual property.

⁶ The UNESCO Recommendation on Open Science (UNESCO, 2021) justifies restrictions to access to scientific knowledge under its definition of open science based on considerations which include the protection of human rights, the protection of intellectual property rights, personal information, and sacred and secret Indigenous knowledge.

6.2.2.4 Agreements related to impact assessment

Similar to a research agreement, an agreement around confidential Indigenous knowledge between and Indigenous community and the Impact Assessment Agency would address the following broad areas (in addition to areas specific to impact assessment) (Impact Assessment Agency of Canada, 2024b):

- Roles and responsibilities
- Sharing confidential Indigenous knowledge with other parties
- Limits on accessing confidential Indigenous knowledge
- How the confidential information will be managed
- The Indigenous community's consent to sharing the confidential Indigenous knowledge
- How the Indigenous knowledge will be used
- The need for a confidentiality agreement, if any

6.2.2.5 Impact assessment confidentiality protections, requirements

In addition to an agreement around confidential Indigenous knowledge, impact assessment processes are subject to legislation and/or have other procedures in place related to the confidentiality of Indigenous knowledge that is shared in the context of an assessment or other formal review process (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024b; Mackenzie Valley Review Board, 2005). For example, any Indigenous knowledge shared by an Indigenous community in confidence under the *Impact Assessment Act* must be treated as confidential (with some exceptions within an impact assessment (e.g., the knowledge is publicly available or it needs to be shared for procedural fairness, natural justice, or for legal proceedings)) (Impact Assessment Agency of Canada, 2024b).

Other jurisdictions also have requirements around confidentiality; for example, in British Columbia, sections of the *Environmental Assessment Act*, 2018, contain protections for Indigenous knowledge to be provided within the environmental assessment process in confidence and the *Freedom of Information and Protection of Privacy Act* also contain provisions to prevent the disclosure of Indigenous knowledge (British Columbia Environmental Assessment Office, 2022). In the Northwest Territories, a *Request for Ruling* can be made and if approved by the Mackenzie Valley Review Board, the traditional knowledge will be protected as confidential (Mackenzie Valley Review Board, 2005). However, detailed confidential Indigenous knowledge may not always be needed. In these cases, information can be summarized or otherwise made more general (Impact Assessment Agency of Canada, 2024b).

6.2.3 Receiving Indigenous knowledge

Be aware of and respect the various needs of the community and of knowledge holders (British Columbia Environmental Assessment Office, 2022). In addition to providing ample advance notice for any engagement (British Columbia Environmental Assessment Office, 2022), explore ideal timing with the community, as Indigenous communities will have season-dependent harvesting

activities such as hunting, fishing, gathering, and trapping during which community members are unavailable for engagement. Materials and information for Indigenous community members should be developed and presented using easy to understand language that avoids technical jargon or complex terminology. Consider if there are also any additional general engagement processes to follow in the community that fall outside culture-specific processes (Impact Assessment Agency of Canada, 2024a). Recognize that an Indigenous community may be dealing with multiple requests to engage, consult, participate in research, and/or share Indigenous knowledge – sometimes the same Indigenous knowledge - at any given time with limited capacity. Consider if there are opportunities to collaborate with other practitioners or proponents to seek and receive Indigenous knowledge while reducing the burden on the Indigenous community.

Engage with diverse knowledge holders and seek Indigenous knowledge in a culturally-appropriate setting that is comfortable for knowledge holders (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a). Engagement settings and methods should be determined on a case-by-case basis, as some topics or issues are best elaborated through alternative means such as storytelling. This could mean engaging knowledge holders through on the land activities and/or in their own language (Impact Assessment Agency of Canada, 2024a). The setting and and/or language could be an important part of the context and meaning of the information (British Columbia Environmental Assessment Office, 2022). A comfortable setting for some Elders or Knowledge holders may include sitting around a bonfire with a Firekeeper, telling one another stories and eating a hot meal. Ensure knowledge holders are able to take the time they need to transmit any stories they wish to share and provide a meal or refreshments if at all possible. Such in-person interactions can support a far richer sharing of knowledge than approaches such as static surveys and virtual meetings.

The proponent should always ask if interpretation or translation would be needed or appreciated by knowledge holders. Interpretation or translation by an individual who has been selected by the Indigenous community may be needed and terms may need to be described where a direct translation does not exist between languages (Impact Assessment Agency of Canada, 2024a). Within settings such as those noted above, participatory video may be an appropriate tool to gather collectively-held Indigenous knowledge and enable community members to document, monitor, and communicate climate change related observations. Participatory video provides an opportunity to co-develop video materials to be shared with the community, and greater audiences if permitted, for use in climate change monitoring and adaptation planning (Signs Of Change: Observations of Climate Change from Samson Cree Nation and Kainai First Nation, n.d.)⁷. If the engagement is recorded (audio, visual, or written), ensure a copy of the community stories shared are made available to the storyteller in an accessible format.

In the context of impact assessment, another approach to receiving Indigenous knowledge could include discussions with the Indigenous community about the possibility of funding the community

⁷ Participatory video was one of the tools used by the Indigenous Climate Change Observation Network (ICCON), which aimed to enable Alberta's Indigenous Knowledge Holders and scientists to co-produce knowledge about climate change-induced changes to support climate change resilience and adaptation (Signs Of Change: Observations of Climate Change from Samson Cree Nation and Kainai First Nation, n.d.).

to carry out an Indigenous knowledge study. Such a study could be completed through a third party that is agreed upon with the community or conducted by the community directly (Impact Assessment Agency of Canada, 2024a).

6.3 Applying Indigenous knowledge in the project development process

6.3.1 Overall approach

When applying Indigenous knowledge within the project development process, technical staff and engineers should make their best attempt to balance the objectives of the project with the relationality involved with Indigenous knowledge (Arsenault et al., 2019). The integration of western science and Indigenous knowledge is more than simply translating Indigenous lived experiences into forms compatible with provincial or federal wildlife management or extracting data from Indigenous communities to input satisfactory information into Canadian assessment frameworks. Indigenous knowledge should not be reduced to facts, observations, or singular practices (Jessen et al., 2021), but rather applied with a holistic approach (Arsenault et al., 2019). For example, a holistic approach would be inclusive of humans and non-humans within the process, as the flora and fauna are active participants within the places they live and should not be viewed as mere subjects reduced to data points (Arsenault et al., 2019). Indigenous communities perceive the environment in a holistic manner, where community members, lands, animals, and waterways are interconnected with one another. Often, western scientific approaches do not align with this all-encompassing view, such as decision-making in resource management relying on the ranking of value placed on a specific species.

The “Two-Eyed Seeing” approach will be discussed in Sections 6.3.4 and 6.3.5, and additional approaches to respectful research are described in the Appendix: Additional approaches to respectful research.

It is important to acknowledge that it can be challenging to adopt a holistic approach while working within western culture and worldviews (Lajoie-O'Malley et al., 2023). Within the impact assessment space, there are calls to address such challenges and those surrounding the process more broadly including through steps such as increased Indigenous power in the decision-making process, strengthened recognition of Indigenous rights, additional funding and opportunities, oversight of Indigenous knowledge by Indigenous peoples, and recognition of fundamental differences between western and Indigenous worldviews and culture (Eckert et al., 2020). Indigenous-led environmental assessment is one specific approach that may help address some of the current short-comings around the ability to meaningfully engage Indigenous knowledge within the existing environmental assessment process (Eckert et al., 2020).

6.3.2 Indigenous knowledge form and context

Context is critical to the appropriate understanding of Indigenous knowledge and Indigenous knowledge should not be taken out of context; stories should not be broken up (British Columbia

Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a; Mackenzie Valley Review Board, 2005). Context can include the community context, the knowledge holder(s), the language of the community, and location (Impact Assessment Agency of Canada, 2024a).

Indigenous community and knowledge holders should be actively involved in presenting and interpreting Indigenous knowledge to make sure the context and meaning of the information remains intact (British Columbia Environmental Assessment Office, 2022; Mackenzie Valley Review Board, 2005). Further, the Indigenous community and knowledge holders must verify any documents or studies they have not written themselves, including summaries of their Indigenous knowledge (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a). Similarly, if existing information such as information found in an existing Indigenous knowledge study will be used for a new project or purpose, ask the Indigenous community for the appropriate context surrounding this information (Impact Assessment Agency of Canada, 2024a).

6.3.3 Analysis

Indigenous knowledge and western science need to be considered within an open, ethical space that allows sharing between the two systems (British Columbia Environmental Assessment Office, 2022; Impact Assessment Agency of Canada, 2024a). Analyze Indigenous knowledge collaboratively with the Indigenous community, examining its relevance while keeping it in full context (Impact Assessment Agency of Canada, 2024a). A workshop or community visit could bring together Indigenous community leadership and project team members including engineers and can produce information that helps with considering Indigenous knowledge for a project (Impact Assessment Agency of Canada, 2024a).

At times, there can be a direct link between Indigenous knowledge and areas of engagement. For example, knowledge of a harvesting area within a proposed project area could suggest potential impacts under an assessment. However, the connections between Indigenous knowledge and areas of interest (e.g., impacts) can be complex and indirect (Impact Assessment Agency of Canada, 2024a).

An example of complex and indirect connections comes from the Mackenzie Valley Review Board (2005). Traditional knowledge holders could identify a link between poor summer physical condition for caribou and greater survival of young trout. The link could be related to an increase in the populations of mosquitoes and black flies that both provide food for juvenile trout and bite caribou. However, if the changes to the caribou and trout are identified separately in a report, this link may be lost.

Aim to gain a stronger understanding of the context around Indigenous knowledge, including linkages between oral histories, culture, and the land, and discuss the information shared (Impact Assessment Agency of Canada, 2024a). Continue to identify linkages as information is shared (Impact Assessment Agency of Canada, 2024a).

When Indigenous knowledge and western science come to different conclusions, investigate the knowledge and views related to both systems (Impact Assessment Agency of Canada, 2024a; Mackenzie Valley Review Board, 2005). For example, western science may anticipate that water levels will decrease with climate change, but the Indigenous community may be experiencing flooding. At the same time, industry may be carrying out clear cutting on the water ways and warming could also be speeding up the rate of snow melt. In this example, it is unclear if impacts on water levels can be attributed to climate change, a single industry, or cumulative impacts from multiple industrial activities. This can also apply to differences in the Indigenous knowledge shared among knowledge holders. For example, Elders may have different knowledge, which can be related to different time periods. Indigenous community members can also identify ecological concerns that require further investigation. In addition, different conclusions can be the trigger for additional research such as the implementation of an adaptive management program or collaborative monitoring (British Columbia Environmental Assessment Office, 2022).

6.3.4 Two-Eyed Seeing

Mi'kmaw Elders Albert Marshall and Murdena Marshall of Eskasoni First Nation are credited for introducing the Two-Eyed Seeing approach (Etuaptmumk) within Canadian research, a concept that involves Indigenous and non-Indigenous researchers "learning to see from one eye with the strengths of Indigenous knowledges and ways of knowing, and from the other eye with the strengths of western knowledges and ways of knowing, and to using both these eyes together, for the benefit of all" (Dimayuga et al., 2020).

The Two-Eyed Seeing approach emphasizes the importance of Indigenous knowledge within the project development process as it is a co-learning journey that encourages the researcher's acknowledgement of needing one another within the relationship and being considerate of their own and one another's values, motivations, knowledges, and actions (Dimayuga et al., 2020). Within Two-Eyed Seeing, both the practitioner and Indigenous knowledge holder are equals (Almack et al., 2023). This approach empowers community members to share Indigenous knowledge through the open involvement of local Elders, knowledge holders, youth, and community members, implementing the participants' lived experiences and personal knowledge in a holistic manner throughout the project development process (Dimayuga et al., 2020).

The Two-Eyed Seeing approach lacks a rigid structure (Almack et al., 2023), and encourages Indigenous and non-Indigenous researchers to value Indigenous knowledge and the culturally-appropriate sharing methods of the community by placing emphasis on community-specific activities, challenging the token inclusion of Indigenous partners and the attempts to absorb Indigenous knowledge into western frameworks, thus making the process more holistic when compared to simply engaging with the community (Dimayuga et al., 2020). As the theory of the Two-Eyed Seeing approach has been described, practical examples of research teams using Two-Eyed Seeing within the engineering sector and/or with potential relevance to climate change and electricity development projects will be described in the next section.

6.3.5 Applying Two-Eyed Seeing

This process of co-learning and co-production is empowered by practitioners and researchers who utilize community-led mechanisms such as ceremonies, talking circles, or semi-structured interviews to receive and integrate influential Indigenous knowledge.

6.3.5.1 Talking Circles & community-led research

Two-Eyed Seeing as a process was demonstrated through a collaboration between a research team and a First Nation in northern Ontario to characterize and monitor impacts of industrial developments and exposure to contaminants related to health impacts (Galatro et al., 2024). The First Nation community was in an area of concern linked to several cases of atypically young acute myeloid leukemia. Although this project was not specific to climate change, the process to seek Indigenous Knowledge in this example may hold insights for climate change projects. The project involved the development of an interdisciplinary framework, which modelled the intersection between quantitative engineering praxes and the First Nation's heritage, identify, and oral stories connecting the land with the physical, mental, emotional, and spiritual wellbeing of families, community, and individuals. This framework and Talking Circles were used to develop the Integrated Knowledge Translation Strategy, which guided the research.

In this project, the Talking Circles allowed the community members and research team to discuss questions and answers, build tools and trust, and share personal or Indigenous knowledge. They were planned according to the Nookiwin Tribal Council's G'minoomaadozimin (We Are Living Well) Protocol of Respectful Interaction. The data collection of Indigenous knowledge within the Talking Circles was prescribed by data release agreements and human ethics protocols aligned with OCAP principles. The Talking Circles were led by Elders and brought together about 20 participants with different characteristics such as sex, gender, and age. These Talking Circles also included cultural practices such as smudging, food sharing/feasting, and locally suitable honoraria for participants.

The initial Talking Circles focused on the development of the Indigenous Integrated Knowledge Translation strategy to guide the project but later Talking Circles focused on the research content. The Circles encouraged relationship building through face-to-face interactions. Ongoing engagement with the First Nation was structured around the Talking Circles to receive, share, and interpret research results in a way for the community to comprehend and provide response. These conversations resulted in the identification of "vital information regarding geoelectrical surveyed lines, locations of contaminated soils, and the positions of previous leachate collection systems" (Galatro et al., 2024).

6.3.5.2 Semi-structured interviews

Two-Eyed Seeing as a flexible process drawing on semi-structured interviews was demonstrated in a research study on environmental changes in boreal landscapes, conducted in partnership with two Quebec-based First Nations (Belisle et al., 2022). In this study, Two-Eyed Seeing was used to assess and compare the perspectives of Indigenous and scientific communities on environmental changes in boreal Quebec, and how collaboration between the perspectives could support

mitigating further degradation. The research team developed two conceptual models to represent the perspectives of either community on a recognized component, with a scientific model informed by research publications and the Indigenous conceptual model developed through semi-structured interviews with local First Nation land-use experts (Belisle et al., 2022).

Semi-structured interviews and participatory mapping was conducted with experienced land users from the respective First Nations. The 23 interviews were conducted in French, English, Cree and Anishinaabemowin, with a member of the research team from one of the communities providing simultaneous interpretation on an as-needed basis. These interviews used close-ended questions and participant self-assessment to verify the land user's level of expertise, examining their attendance on the land and how active they were with any of the six landscape practices of moose hunting, trapping, fishing, goose hunting, education, and restoration (Belisle et al., 2022).

The interview transcripts were thematically analyzed and together with the participatory mapping described below, the results informed the conceptual model structure. Excerpts were anonymized and the research team met with each First Nation to discuss and validate the coding accuracy, seeking further interpretation and confirming ambiguous statements or important phenomena derived from anecdotal events (Belisle et al., 2022).

Using Two-Eyed Seeing, the research team found and examined areas of convergence and areas of divergence between the models (Belisle et al., 2022). The influence of climate change on boreal landscapes was present within both the Indigenous and scientific models, with the primary influence of forestry as a driver. The Indigenous model found climate change to be associated with wildlife distribution and weather, and the scientific model found climate change to be associated with wildfire activity, weather and forest productivity, and tree species distributions. Areas of divergence included mining and hydroelectric development being two drivers exclusive to the Indigenous model, while wildfires were more prominent in the scientific model. The divergences between Indigenous and western scientific knowledge demonstrated the ways each knowledge system can complement the other through the bridging of gaps in qualitative or quantitative data.

Another example of the use of interviews when applying a Two-Eyed Seeing approach, comes from a research study conducted in the Saskatchewan River Delta (Abu et al., 2020). In the study, oral history interviews were held with Elders to receive first-hand knowledge of key historical events to fill gaps in archival data, and 42 semi-structured interviews were held with fishers, trappers, hunters, and plant harvesters to receive their perspectives on changes to the environment and related land-based activities. Interview guides were developed in collaboration with the Indigenous community partners, with the interviews ranged from 45 minutes to four hours. The interviews were audio recorded and transcribed verbatim before being analyzed and the results were presented to the community for verification. The results of the interviews were examined alongside instrumental and archival records (western science). Researchers found that the Indigenous knowledge and western science often corroborated findings but there were instances where Indigenous Knowledge provided information that was not available through other sources. For example, Indigenous knowledge holders were able to speak to changes in the duration of berry-growing seasons, and in the growth patterns of some plants. The Indigenous knowledge and

qualitative archival records also identified local impacts from changes that were not detected through instrumental observations. There were also instances where the knowledge systems did not agree, and which identified the need for further. For example, Indigenous knowledge indicated pike were thriving, but the instrumental observations indicated the pike commercial harvest had nearly ceased; this difference may reflect a decline in the commercial harvest of pike.

A separate research study in the Northwest Territories bridged Indigenous knowledge with western scientific knowledge using interviews with Elders, field data, existing models, and expert judgements to address questions about ecosystem health in the Slave River Delta with a holistic understanding (Mantyka-Pringle et al., 2017). Researchers used one-on-one interviews with community Elders to receive Indigenous knowledge regarding the region's social-ecological changes over a certain period, and to compare current experiences within the Slave River Delta region in relation to the past. The research team used the Elder responses, field observations, and document reviews to develop a "Two-Eyed Seeing" Bayesian Belief Network designed to bridge Indigenous knowledge and western scientific knowledge to understand the cumulative environmental impacts of stressors on the Slave River Delta ecosystem. The model framework for the Bayesian Belief Network was presented in two workshops to a group of experts comprised of Indigenous harvesters and Elders, and non-Indigenous government staff and scientists, to confirm and populate the remaining indicators and their respective importance.

6.3.5.3 Participatory mapping

Participatory mapping is another approach that has been used when applying Two-Eyed Seeing, in the Boreal Quebec study from Belisle et al. (2022), the Indigenous land experts were also invited to indicate places or perimeters of areas considered high value or low value on a laminated map of the region, facilitating deeper discussions on usage of and experiences on the land (Belisle et al., 2022). This participatory mapping supplemented the findings from the semi-structured interviews.

The use of participatory mapping exercises specific to a community's connection to the land and interviews were further illustrated in a study based on the Central Coast of BC, in which the research team examined zooarchaeological, historical, Indigenous, and western science knowledge to better understand changes in Canada's northern abalone social-ecological system (SES) (Lee et al., 2018). Researchers invited Heiltsuk abalone experts to participate in semi-structured interviews where they were asked to reconstruct the number, size, and spatial distribution within a specific site over each decade. The Heiltsuk abalone experts were then asked to place colour-coded, length-classified abalone shells on a table, representing abalone densities within a specified area that the research team quantified and photographed (to view the photographs of abalone abundance and size reconstructions based on Indigenous knowledge, please see Figure 3) (Lee et al., 2018). The BC Central Coast study conducted integrated modeling of Indigenous knowledge and western science data to obtain a broader understanding of recent changes and quantify uncertainty in both sets of knowledge. The study found Indigenous knowledge was crucial to understanding the baseline shift in abalone population.

These examples demonstrate that in addition to culturally-based techniques such as Talking Circles, the application of Two-Eyed Seeing can also be supported by other mechanisms such as semi-structured interviews and participatory mapping.

7. Looking Forward

This document presents a starting point for guidance around seeking, receiving, and applying Indigenous knowledge based on promising practices from the grey and peer-reviewed literature. Indigenous knowledge is comprised of and provides intergenerational practices and teachings based in the local history and environment, allowing more accurate and community-specific measurements of long-term ecological change. Input from Indigenous communities, knowledge holders, and practitioners as the guidance is applied will be key to ensuring the content remains relevant, respectful, and supports mutually beneficial collaboration around electricity sector projects. Indigenous knowledge allows for a holistic approach to planning and operations of projects, as community-based inputs such as water vitality or animal ethics measure qualitative data based in lived experiences and may identify links between areas of the environment seemingly non-related.

Areas of related practice such as Indigenous engagement, Indigenous research, and environmental/impact assessment will also continue to evolve. Approaches such as Indigenous-led environmental assessment is one potential approach that could help address current challenges related to Indigenous knowledge in the context of impact assessment. These developments and further examples of related projects in the context of climate change can support enhancements to this guidance document into the future.

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A. Appendix: Additional approaches to respectful research

The following sections describe additional respectful research approaches. While these do not specifically focus on weaving together Indigenous knowledge and western science, practitioners may benefit from an awareness of these approaches to research with Indigenous communities.

I. Research is Ceremony

“Research is Ceremony” describes an approach to Indigenist research brought forward by Shawn Wilson and other Indigenous scholars. Under this approach, the researcher and participant create a new collective understanding through a sacred ceremonial process that rests on Indigenous worldviews. The phrase “All our relations” is spoken at the beginning and end of ceremonies and reflects the core idea of relationality – the connection between all that is living and non-living, as well past and future generations, and the connection between multiple realities and knowledge systems. Rather than a focus on objectivity, researchers and participants are intertwined and the research is shaped by the researcher’s own relationship with the living, non-living, and ideas surrounding them. For Wilson, the researcher is also accountable for the impact of their research on the many components within this concept of relationality. Non-Indigenous researchers may undertake “Research is Ceremony” but the approach must be honoured and well articulated (Dimayuga et al., 2020).

II. Doing Research in a “Good Way”

Doing research in a “Good Way” focuses on conducting western empirical research in a way that is culturally and ethically appropriate, in alignment with the principles of respect, relevance, reciprocity, and responsibility. Another approach that complements these principles of respectful research is Participatory Action Research (PAR), which aims to create positive social change for research. Researchers can also use research methods which are based on an Indigenous worldview to conduct respectful research. Examples include Indigenist research methodologies that include individual storytelling and collective sharing (Dimayuga et al., 2020).

